

KALLE HEINONEN

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PERSONAL DETAILS

I am a 25 yearold MSc in mathematics. I graduated from the study track of geometry, algebra and topology at the University of Helsinki. My master's thesis was on the Poincaré-Hopf index theorem, which relates data from isolated singularities of vector fields on closed manifolds to it's purely topological invariants. Specifcly a proof utilizing de Rham theory and Morse theory.

EDUCATION

Bachelor of Science in Mathematics, University of Helsinki Sep 2020 - May 2023

Minor in computer science

Master of Science in Mathematics, University of Helsinki Sep 2023 - April 2025

SKILLS

Technical Skills	Problem solving, LaTeX, Python, Haskell, basics of Matlab, R, Excel and SQL.
Mathematics knowledge	https://kh-xyz.github.io/mathlist ← elaborated here
Computer science knowledge	The basic minor studies such as python programming, computer organization, algorithms and databases. Additionally functional programming in Haskell.
Languages	Finnish: Native, English: Full professional proficiency, Swedish: ~B1.

JOB EXPERIENCE

- Non-military service at Aalto univerty Department of Mathematics and System's analysis	Sep 2025- Aug 2026
- Live musician - Freelance	Sep 2021 - Current
- High school mathmematics / computer science substitute teacher - Freelance	Sep 2021 - Mar 2025
- Teaching assistant at University of Helsinki & Aalto university - Seasonal	Oct 2021 - May 2026
- Villa host at Hämmäläisten Ylioppilassäätiö - Full-time	Aug 2026

OTHER RESPONSIBILITIES

- Board member and other responsibilities at Matrix ry	Jan 2021 - May 2025
- Vice chair and other activities at Moodi ry	Jan 2024 - Dec 2026
- Board and other activities at Scout organization Vuosaaren Vesipääskyt ry	Sep 2008 - Mar 2024
- Student member of the board of the Master's program in mathematics and statistics and Bachelor's program in mathematical sciences	Jan 2023 - Jun 2025
- Multiple responsibilities at Hämmäläis-osakunta student nation	Jan 2021 - Dec 2025
- Vice representative in the representative council at Helsingin yliopiston ylioppilaskunta	Jan 2023 - Dec 2024
- Vice artistic director at Polirytmii ry	Jan 2026 - Dec 2026

TEACHING EXPERIENCE

Discrete mathematics (5 ETC), Algebraic structures I-II (5 + 5 ETC), Topology IA-IB 5 + 5 (5 ETC) (twice both courses), Academic skills in mathematics (2 ETC), Complex analysis (5 ETC), Functional analysis I-II (5 + 5 ETC), Introduction probability and statistics (5 ETC) (twice).